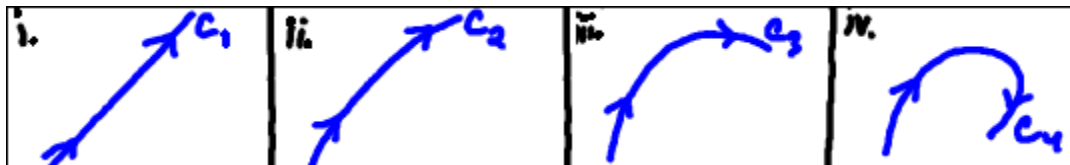


2.5 Curvature

We will now explore the notion of curvature. We begin by reviewing some terminology.

Definition: A parameterization $\mathbf{r}(t)$ is called smooth on an interval I if $\mathbf{r}'$ is continuous and $\mathbf{r}'(t) \neq \mathbf{0}$. A curve is smooth if it has a smooth parameterization. A smooth curve has no sharp corners or cusps.

Remark: For a smooth curve given by $\mathbf{r}(t)$ when the unit tangent vector, $\mathbf{T}(t)$, turns it does so continuously. When a smooth curve has sharp turns $\mathbf{T}(t)$ will turn quickly. When a curve turns quite slowly, $\mathbf{T}(t)$ will not turn quickly. Consider the pictures below for examples of this idea.



This leads to the following definition.

Definition: The curvature of a smooth curve is $\kappa = \left\| \frac{d\mathbf{T}}{ds} \right\|$.

Theorem: The curvature of a smooth curve given by $\mathbf{r}(t)$ can be expressed as

$$1) \quad \kappa(t) = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|}.$$

OR

$$2) \quad \kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}$$

Proof: We will prove (2). A proof of (1) is available in Blackboard.

Note: Part (b) of this theorem will tend to be more useful since it only requires that we know the vector-valued function $\mathbf{r}(t)$, we don't need to compute the tangent vector $\mathbf{T}(t)$, and we can easily change the parameter if needed.

Example 1: Consider the circle given by $\mathbf{r}(t) = \langle a \cos(t), a \sin(t) \rangle$ where $0 \leq t < 2\pi$ and $a > 0$. Find the curvature of this circle.

Example 2: Let C be the ellipse $x^2 + 16y^2 = 16$ oriented counterclockwise. Measure the curvature at each endpoint of its major and minor axes.